

Locating street markets: analysis from qualitative and quantitative factors using a discrete choice model, mixed integer nonlinear programming and fuzzy AHP

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ARTICLE INFO	ABSTRACT
<i>Article history:</i> Received: Received in revised format: Accepted: Available online: <i>Keywords:</i> Street Markets Facility Location Urban Planning Fuzzy AHP Discrete Choice Model	Street markets can play a key role in addressing food insecurity and promoting equitable access to nutritious food, particularly in urban areas where food deserts are common. The strategic location of street markets can enhance their sustainability and positive impact on urban communities. Despite their potential, previous studies have rarely integrated urban planning constraints with mathematical optimization models for street market locations. This gap limits the ability to effectively combine qualitative and quantitative factors in street market planning. This paper presents a mathematical model combining Mixed Integer Non-Linear Programming (MINLP) and Fuzzy Analytic Hierarchy Process (Fuzzy AHP) to determine the optimal locations for street markets. The objective is to integrate demand, distance, urban regulations, and community needs into a decision-making framework. The model was applied to the district of Bosa in Bogotá, Colombia, using geospatial data and urban planning criteria. Factors such as population density, road infrastructure, and zoning regulations were considered. The Fuzzy AHP incorporated urban planning criteria drawn from the literature and validated through expert judgment, while the MINLP addressed real-world constraints such as buffer zones and customer accessibility. Results show that incorporating qualitative urban criteria via Fuzzy AHP yields more spatially balanced and accessible market locations, albeit at the cost of slightly lower demand capture than purely demand-driven solutions. Scenario analysis further indicates that the model is robust to variations in walking-distance thresholds, the number of markets, and the temporal distribution of demand.

1. Introduction

Street markets, with a history spanning centuries around the world, are more than just sources of fresh produce and various goods; They are an integral part of the social fabric in many parts of the world. Despite facing political decisions and competition from different retailers over the years, these markets maintain their importance in both developed and developing countries. These markets are typically run by local vendors who set up and dismantle their stalls in a single day, offering a wide range of products, including some that are rare in conventional

outlets such as corner stores and supermarkets. In addition, as these street markets are served by local producers and do not rely on intermediation, they can offer lower prices in comparison with their competitors. These markets are marked by a high level of activity, which can explain why they are preferred by many customers for buying fresh foods (Costa et al., 2013; Nogueira et al., 2018).

The contribution of street markets to food security strongly depends on their location, product availability, and accessibility. However, the problem of locating street markets, also referred to as farmers' markets, open-air markets, or mobile markets, remains relatively underexplored in academic literature, despite evidence of their positive impact on food access and nutrition (Robinson et al., 2016; Jilcott Pitts et al., 2014). Inadequate location strategies, not only for street markets but also for other fresh food retailers, can contribute to the emergence of so-called "food deserts," whereas well-designed location decisions can improve supply chain performance, enhance market sustainability, and promote equitable access to nutritious food (OBSSAN, 2018a; OBSSAN, 2018b). In practice, street markets coexist with supermarkets and other retailers within a broader urban food ecosystem, making their spatial integration particularly relevant (Morón, 2006; Orjuela-Castro et al., 2017).

The relationship between street markets and urban planning is therefore critical. Traditionally, their locations have been defined by public authorities based on empirical criteria and political considerations. This approach often neglects key spatial and functional aspects such as road accessibility, traffic conditions, land use, population density, and urban design, all of which directly affect market performance and accessibility (Rajagopal, 2010). From a facility location perspective, the problem can be formulated as identifying market sites that maximize demand capture or market share, a goal typically addressed through mathematical optimization models incorporating discrete choice behavior (Train, 2009). These models rely on measures of customer attractiveness reflecting perceived utility, influenced by factors such as distance, prices, assortment, technological adoption, and visual appeal. Despite their potential, formal location models and urban planning considerations have rarely been integrated in a unified analytical framework.

Despite the extensive literature on facility location and market siting, existing Mixed Integer Non-Linear Programming (MINLP) and Capacitated Facility Location Problem (CFLP) based approaches generally model markets as permanent and homogeneous facilities, with limited consideration of the specific characteristics of street markets as temporary, mobile, and highly regulated users of public space. At the same time, urban planning studies on street markets and informal food systems tend to focus on governance, regulation, and social impacts, but rarely provide decision-support tools capable of systematically integrating these aspects into location decisions. This disconnect reveals a clear gap between operations research models and urban planning perspectives in the context of street market location. This paper addresses this gap by proposing a MINL model combined with a Fuzzy Analytic Hierarchy Process (Fuzzy AHP) to jointly capture quantitative performance measures and qualitative planning criteria. The contribution of this study is threefold. First, it formulates a street market location problem that explicitly accounts for the temporary nature of markets and for urban and regulatory constraints associated with the governance of public space. Second, it integrates optimization modeling with Fuzzy AHP to incorporate literature-based qualitative criteria, validated through expert judgment, alongside demand-driven objectives. Third, it presents an applied case study in a municipality located in the southwestern area of Bogotá (Colombia), illustrating how the proposed approach can support public authorities in designing transparent, data-driven, and context-sensitive location policies for street markets within complex urban food systems.

This paper is organized as follows: Section 2 introduces the context of the study, providing insights into Bogotá's urban aspects and the characteristics of the district of Bosa. Section 3 presents the theoretical framework, explaining the methodologies of discrete choice models, the Analytic Hierarchy Process (AHP), and fuzzy set theory. Section 4 outlines the methodological framework, including data preparation and evaluation of potential locations. Section 5 shows the optimization model. Section 6 provides computational results and analysis. Finally, Section 7 summarizes the conclusions and highlights directions for future research.

2. Context

Bogotá, the capital of Colombia, has around 8 million inhabitants according to the last census in 2018 (DANE, 2021) from which around 25.4% and 4.2% of the population is respectively moderately and severely

food insecure (DANE, 2023). One of the most vulnerable districts is Bosa, situated within the city of Bogotá, Colombia. Bosa represents the 7th administrative division of Bogotá, predominantly inhabited by low and lower-middle-class residents. The total population of this region sits at approximately 823,000 inhabitants (DANE, 2018) and has an area of 2466 hectares (Alcaldía Local de Bosa, n.d.). The density of around 33,400 inhabitants per square kilometer is extremely high.

Bosa borders three other districts of Bogotá. The topography of Bosa has a flat terrain formed by alluvial deposits from the Bogotá and Tunjuelo rivers. The Tunjuelo river basin supports about two and a half million low-income residents. This area faces significant environmental impacts from landfill and mining activities, contributing to socio-spatial segregation. The western part of Bosa, which is flat and low-lying, has been experiencing rapid growth and development.

The municipality of Bosa has severe problems with food security according to governmental sources (Veeduría Distrital, 2024). The area has very few supermarkets, open street markets and other sources of nutritious food. In addition, most people in the municipality do not have cars and depend on public transportation (or foot) for their shopping.

Due to its extremely high population density, the construction of market facilities is virtually impossible as of now. Thus, a strategy to locate street markets would rely exclusively on open areas such as public parks. Bosa has 244 public parks that are part of the District Park System: two of them are metropolitan parks, 139 neighborhood parks, 96 pocket parks and 7 zonal parks, which have specialized equipment such as sports centers, swimming pools, courts and skating rinks (Alcaldía Mayor de Bogotá, 2016).

The street markets in Bogotá are an initiative led by the Secretariat of Economic Development of Bogotá (Secretaría Distrital de Desarrollo Económico, SDDE in Spanish) aimed at consolidating marketing channels for small-scale food producers and processors, primarily from the central region and municipalities of Cundinamarca. According to the SDDE, this initiative focuses on agroecological production and the rural economy, fostering regional integration, associativity, and the exchange of ancestral knowledge. Generally located in parks in the northern and eastern parts of Bogotá, the street markets aim at good agricultural practices and food security while promoting social inclusion by offering business opportunities to small farmers (SDDE, n.d.). The operation of the street markets is heavily subsidized by the SDDE and at the time of writing, it is not sustainable. As of today, less than 2% of the food in Bogotá is commercialized through these markets, according to our sources.

Street markets may play a role in food security by facilitating the distribution, availability, and accessibility of food. They connect producers, distributors, and consumers, influencing prices and supply dynamics (Timmer, 2017). In many regions, informal markets are the primary source of food for low-income populations (Battersby & Watson, 2018). These markets not only support livelihoods but also contribute to social cohesion by serving as spaces for community interaction (Ghanem, 2008). These markets present urban restrictions, such as zoning laws, infrastructure limitations, and regulatory barriers, often disrupt the operation of food markets. City planning decisions that overlook food distribution systems can create food deserts, areas where access to affordable and nutritious food is severely limited (Pothukuchi & Kaufman, 1999). Informal street markets often lack basic amenities like clean water, electricity, and waste disposal services, contributing to health and sanitation challenges. Vendors operating in crowded public spaces can exacerbate spatial conflicts with motorists, pedestrians, and formal businesses, leading to increased tensions (Ojeda & Pino, 2019). However, despite their importance, there is limited academic literature that deeply explores how markets impact food security, particularly in urban settings.

2.1 Facility location problem for street markets

The street market location problem is a variant of the well-known Facility Location Problem (FLP). The Facility Location Problem (FLP) is a classic optimization problem in operations research and logistics that involves determining the optimal locations for one or more candidate facilities (such as warehouses, factories, distribution centers, or service stations) that serve a set of customers. The objective is to select a subset of these candidate facilities, so to optimize one or more criteria, such as minimizing costs, maximizing accessibility, or capturing demand (Owen & Daskin, 1998). There are various FLP variants, including set covering, maximal covering, p-center, p-dispersion, p-median, and fixed charge, among others (Daskin, 2008; Drezner & Hamacher,

2002). Most of these problems are NP-Hard. Over time, traditional models have evolved to incorporate more complex factors that better represent real-world conditions (Farahani et al., 2019; Pliego-Marugán et al., 2021). A comprehensive taxonomy of these problems can be found in Farahani et al. (2010).

In many cases, the decision-maker has control over which customers can serve; however, in other instances, the customer's behavior cannot be controlled, and it is influenced by competitors. This leads to the Competitive Facility Location Problem (CFLP). In the CFLP, the most common objective is to maximize market share, often competing directly with others for the same pool of customers (Aboolian et al., 2007). Demand in such environments becomes dynamic and dependent on factors such as the attractiveness, distance, pricing strategies, and service quality of the current and proposed sites (Biesinger et al., 2015).

There are different modeling and solution methods for FLP, often based on exact methods such as Mixed Integer Linear Programming (MILP) (Mann, 2022; Ralphs et al., 2018), Branch and Bound (Erlenkotter, 1978) or Dynamic programming (Shah & Farach-Colton, 2002). Various heuristic and metaheuristic techniques have also been proposed to handle larger or more complex problems. Examples that include Tabu Search (Glover, 1997), GRASP (Greedy Randomized Adaptive Search Procedure) (Resende & Ribeiro, 2014), Genetic Algorithms (Bhattacharjee & Mukhopadhyay, 2023), Simulated Annealing (Mavridou & Pardalos, 1997), and Ant Colony Optimization (Misra & Chakraborty, 2024), have also been successfully applied to many location problems (Basu et al., 2014; Fischetti & Lodi, 2011; Shen et al., 2011).

The FLP studies in Operations Research have generally concentrated on quantitative factors that in many cases do not capture the complexity of the decision making, especially in public facilities; there is a growing trend toward incorporating qualitative factors into the decision-making process. In Lee et al. (1981), the authors integrated integer goal programming with qualitative factors in the location of potential production plants where the focal firm is assessing six potential sites across four states. Later, Arinze and Banerjee (1992) developed a knowledge-based system to address qualitative aspects using user-defined heuristics in location analysis in the context of manufacturing firms seeking to start new businesses or to expand or modify the scope of their current operations. In Yang and Lee (1997), the study introduced an AHP decision model for facility location with qualitative and quantitative factors for the location of a new facility or a relocation of an existing one. Other studies also used AHP: for instance, Badri (1999) combined AHP and multi-objective goal programming to address conflicting factors for facility location-allocation decisions in an international setting. Using a similar approach, Chuang (2001) combined AHP and Quality Function Deployment (QFD) to support facility location decisions and the work of Partovi (2006) proposed a solution to the facility location problem by integrating both external (customer demands, market conditions) and internal criteria (key processes within the manufacturing organization) with AHP, QFD and analytic network process (ANP). In Kahraman et al. (2010), the authors presented various approaches to the location selection problem, focusing on fuzzy multi-criteria decision-making methods, particularly fuzzy AHP and fuzzy TOPSIS (Technique for Order Preference by Similarity to Ideal Solution). Another paper that uses fuzzy AHP is the work of Ozgen and Gulsun (2014), who combined multi-objective possibilistic linear programming MOPLP and fuzzy AHP to minimize the total transportation and facilities costs as first objective and maximize the total qualitative factor benefits for facility location for the second objective. Then, Kim and Hwang (2018) applied spatial and multi-criteria analysis using GIS (Georeferenced Information System) and AHP to evaluate quantitative and qualitative factors for selecting a sustainable facility site. Most recently, Şekerci and Aydin (2022) applied Fuzzy AHP and stochastic mixed-integer programming for facility location and network design under uncertainty for a bottled water production-distribution system. The objective function was the minimization of the total cost (production, warehouses, transportation, etc.).

Street markets are increasingly integrated into urban development strategies, particularly in the context of improving food distribution networks through models that include hubs and street markets (Zazo-Moratalla & Orellana-McBride, 2024). Regarding street markets, strategy studies have traditionally focused on cultural and social aspects (Siegnier et al., 2018). Among the few studies that use mathematical models, Tong et al. (2012) developed a MILP model for the location of street markets addressed spatial and temporal constraints in the objective function was to minimize the overall travel distance. Later, although not specifically on street markets, Zambrano-Rey et al. (2019) added complexity in a model for retail stores with the incorporation of probabilistic

functions such as the well-known "Logit", using factors such as price and distance to define these probabilities. More recently, Mejía et al. (2023) introduced a MINLP (CFLP) model for the location of street markets with location-dependent demand. The study was carried out with data from Valparaíso, Chile. A summary of the literature review is shown in Table 1.

Table 1. Summary of literature review.

Refer- ence	Method/Model	Application	Key Findings
Lee et al. (1981)	Integer Goal Program- ming + Qualitative Fac- tors	Facility location for po- tential production plants	Integrated qualitative fac- tors in facility location analysis across six sites.
Arinze & Banerjee (1992)	Knowledge-Based Sys- tem (Heuristics)	Manufacturing firms' lo- cation decisions	Used user-defined heuris- tics to address qualitative aspects of location analy- sis.
Yang & Lee (1997)	AHP Decision Model	Facility location/reloca- tion	Combined qualitative and quantitative factors for location decision-mak- ing.
Badri (1999)	AHP + Multi-Objective Goal Programming	International facility lo- cation-allocation deci- sions	Addressed conflicting factors in international contexts.
Chuang (2001)	AHP + Quality Function Deployment (QFD)	Facility location deci- sions	Supported decisions by integrating customer de- mands and internal pro- cesses.
Partovi (2006)	AHP + QFD + ANP	Facility location in manu- facturing	Integrated external (mar- ket) and internal (pro- cess) criteria.
Kahraman et al. (2010)	Fuzzy AHP + Fuzzy TOPSIS	Location selection with MCDM	Applied fuzzy MCDM methods to address un- certainty in location deci- sions.
Ozgen & Gulsun (2014)	Fuzzy AHP + MOPLP	Facility location for transportation costs	Minimized costs and maximized qualitative benefits.
Kim & Hwang (2018)	GIS + AHP	Sustainable facility site selection	Used spatial and multi- criteria analysis for loca- tion decisions.
Şekerci & Aydin (2022)	Fuzzy AHP + Stochastic Mixed-Integer Program- ming	Bottled water production- distribution system	Modeled network design is under uncertainty to minimize total costs.
Zazo- Moratalla & Orel- lana- McBride (2024)	Urban Development Strategies	Food distribution net- works with markets	Highlighted the integra- tion of markets in urban development strategies.
Siegner et al. (2018)	Cultural/Social Focus	Street markets	Focused on social and cultural aspects without mathematical modeling.
Tong et al. (2012)	MILP Model	Street market location	Minimized travel dis- tances with spatial and temporal constraints.
Zam- brano- Rey et al. (2019)	Probabilistic Model (Logit)	Retail store location	Incorporated probabilistic factors like price and dis- tance.

Mejía et al. (2023)	MINLP (CFLP) Model	Street markets with location-dependent demand	Modeled demand based on location data from Valparaíso, Chile.
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Previous studies in mathematical programming always consider a set of candidates which, in theory, comply with all urban constraints. However, not all candidates are equal in terms of preference by urban planners. Aspects such as parking areas for loading and unloading, availability of public services and interrelations with other aspects of urban life are often neglected. Considering these aspects could help address urban planning needs and potentially enhance market performance. This gap underscores the need for a more comprehensive approach. To address this, we developed a mathematical model that incorporates these qualitative and quantitative aspects through Mixed Integer Non-Linear Programming (MINLP) and fuzzy AHP.

Taken together, the reviewed studies reveal common limitations. Most facility location and competitive location models assume that candidate sites are already feasible from an urban planning perspective, treating regulatory and spatial suitability as exogenous inputs. Conversely, multi-criteria approach such as AHP or Fuzzy AHP are often used as standalone ranking tools, rather than being fully integrated into demand-driven optimization models. In addition, most existing works focus on permanent facilities, with limited consideration of the operational and regulatory characteristics of temporary street markets.

This study addresses these gaps by proposing an integrated framework that embeds qualitative urban suitability, captured via Fuzzy AHP, directly into a discrete-choice-based optimization model. In doing so, it advances existing approaches by jointly considering customer behavior, urban planning constraints, and the nature of street markets within a single and integrated decision-support formulation

3. Theoretical framework

This section outlines the theoretical foundations of the proposed modeling framework for the location of street markets, emphasizing how behavioral, qualitative, and optimization-based approaches are jointly employed to address the problem. Customer demand is represented through a discrete choice model, which captures how individuals allocate their purchases among competing food outlets based on perceived utility. In parallel, qualitative urban and infrastructural factors that influence the suitability of potential locations are assessed through the Analytic Hierarchy Process, extended with fuzzy set theory to account for uncertainty and subjectivity in expert evaluations. These components are subsequently integrated into a mixed-integer nonlinear programming model, where demand allocation, urban planning constraints, and operational considerations are simultaneously considered. The combined use of these methods enables a comprehensive analysis that links individual choice behavior with urban suitability and system-level decision-making.

3.1 Discrete choice model: utility function

Discrete choice models are used to predict a selection between a set of mutually exclusive discrete alternatives (Train, 2009). These models are widely used in various fields such as economics, marketing, transportation, and public policy to understand and predict individual decision-making behavior. The underlying idea behind discrete choice models is that an individual j selects the alternative i that provides him/her with the highest utility. Let U_{ij} be the utility perceived by individual j of alternative i . This utility is defined as follows: $U_{ij} = V_{ij} + \varepsilon_{ij}$, where V_{ij} corresponds to the observed part of the utility and ε_{ij} is a random error term or unobserved part. The observed part of the utility is generally expressed as a weighted sum of K attribute values. These attributes are the factors that affect the choice and that can be observed.

$$V_{ij} = \beta_{0i} + \sum_{k=1}^K \beta_k v_{ijk} \quad (1)$$

Where β_{0i} is the alternative specific constant, β_k is the weight of the k^{th} attribute and v_{ijk} is the value of the k^{th} attribute given by individual j to alternative i , generally collected from surveys or interviews. The attributes can be distance, price, convenience, etc. The values of the weights must be estimated from the data. Numerical methods are normally used for this estimation (Zhao et al., 2018).

If the random error term follows a Gumbel distribution, then the probability PR_{ij} of individual j of selecting alternative i is given by:

$$PR_{ij} = \frac{e^{v_{ij}}}{\sum_{l \in C} e^{v_{lj}}} \quad (2)$$

This equation is the classical multinomial logit formula (Train, 2009). In this research, the discrete choice model is incorporated into the MINLP.

3.2 Analytic hierarchy process (AHP)

The analytic hierarchy process (AHP) breaks down a complex decision into a hierarchy of sub-levels, each assessed not only for its impact but also for its interrelationships with other levels (Taherdoost, 2017). The hierarchy typically consists of three levels:

1. Goal: The overall objective of the decision.
2. Criteria: The factors or attributes that influence the decision.
3. Alternatives: The possible options or choices available.

The AHP includes the following steps:

1. Pairwise comparison: the elements at each level of the hierarchy are pairwise compared in terms of their impact on an element above them. For example, at level 3, all alternatives are compared in terms of each of the criteria specified at level 2. This involves making judgements about the relative importance of each element with respect to the others. Normally the judgements are made on a 1-9 scale (Vaidya & Kumar, 2006).
2. Priority calculation: The comparisons are used to create matrixes at each level of the hierarchy. The eigenvector method is then applied to those matrixes to derive priority weights, which indicate the relative importance of each element (Vaidya & Kumar, 2006).
3. Consistency check: AHP includes a consistency index (CI) and consistency ratio (CR) to ensure that the judgments made during pairwise comparisons are consistent and coherent, as indicated by expert opinions. If the CR is below a certain threshold (usually 0.1), the level of consistency is acceptable.
4. Synthesis: The final step involves aggregating the priority weights across the hierarchy to determine the overall priority of each alternative. This synthesis provides a quantitative basis for making the final decision, effectively calibrating a numeric scale for measuring both quantitative and qualitative performances (Vaidya & Kumar, 2006).

3.3 Fuzzy set theory

The theory of fuzzy sets, introduced by Zadeh in the 1960s, extends the concept of classical sets by allowing elements to have degrees of membership (Zadeh, 1978). Instead of being limited to either belonging or not belonging to a set, as in classical theory, elements in a fuzzy set can belong partially. This approach is especially useful in situations where the boundaries between categories are not clear, such as in complex decision-making processes. For example, terms like "tall" or "short," or "near" and "far," can be expressed more precisely using degrees of membership, better reflecting the inherent ambiguity of these concepts (Lee, 2014).

Membership functions allow for measuring the degree to which an element belongs to a set. These functions assign a value between 0 and 1, where 0 means that the element does not belong to the set, and 1 indicates full membership. The most common forms of these functions include triangular, trapezoidal, and Gaussian types (Fu et al., 2023). For example, a triangular membership function defines a linear ascending and descending

relationship in values, providing a simple way to model partial membership in fuzzy terms. This triangular membership function is represented in Figure 1 and the formula:

$$\mu_{\tilde{A}}(x) = \begin{cases} \frac{x-l}{m-l} & \text{for } l \leq x \leq m \\ \frac{u-x}{u-m} & \text{for } m \leq x \leq u \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where l , m and u , represent the lower, middle, and upper values respectively of the membership function. Thus, the fuzzy set can express a range of possible values with soft boundaries instead of precise, crisp values (Buckley, 1985).

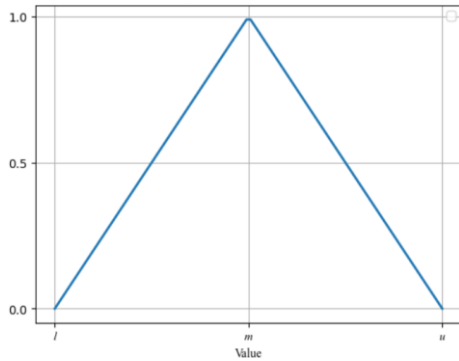


Figure 1. Triangular membership function.

3.3.1 Operations with triangular fuzzy numbers

The following arithmetical operations are used for performing calculations with triangular fuzzy numbers, which are a common representation in fuzzy theory. These operations (addition, multiplication, and inverse) are needed to combine, compare, and evaluate the alternatives and criteria in the Fuzzy AHP approach (Chang, 1996). Let $\tilde{A}_1 = (l_1, m_1, u_1)$ and $\tilde{A}_2 = (l_2, m_2, u_2)$ be two triangular fuzzy numbers and the basic operations are:

$$\text{Addition } \tilde{A}_1 \oplus \tilde{A}_2 = (l_1 + l_2, m_1 + m_2, u_1 + u_2) \quad (4)$$

$$\text{Multiplication } \tilde{A}_1 \otimes \tilde{A}_2 = (l_1 \times l_2, m_1 \times m_2, u_1 \times u_2) \quad (5)$$

$$\text{Inverse } \tilde{A}^{-1} = \left(\frac{1}{u}, \frac{1}{m}, \frac{1}{l} \right) \quad (6)$$

$$\text{Lower limit } l, m, u > 0 \quad (7)$$

3.3.2 Fuzzy AHP

Fuzzy AHP is a decision-making technique that combines hierarchical analytical methods with fuzzy logic. This approach, initially proposed by Buckley (1985), handles the uncertainty and imprecision typical in the evaluation process. The phases are:

1. Hierarchical structuring: First, the problem is organized into a hierarchical structure, which includes a main goal, a predefined number of criteria, and the alternatives as is shown in Figure 2 – Section A. The elements of the hierarchical structure are matrices with pairwise comparisons.
2. Establish the value of all performance indicators corresponding to each alternative and each criterion.

3. Make pairwise comparisons between criteria using Triangular Fuzzy Numbers (TFN) instead of crisp numbers as in the traditional AHP. Establish the fuzzy comparison matrices for all criteria. In a fuzzy comparison matrix $\tilde{M}$ each element of the matrix represents the relative importance of one criterion over another using TFNs.

$$\tilde{M} = \begin{bmatrix} 1 & \tilde{m}_{12} & \dots & \tilde{m}_{1j} \\ \tilde{m}_{21} & \ddots & \dots & \tilde{m}_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{m}_{i1} & \tilde{m}_{i2} & \dots & 1 \end{bmatrix} \quad (8)$$

Where $\tilde{m}_{ij}$ the preference of the i – th criterion over j – th criterion for the decision maker.

4. Weight calculation as shown in Figure 2 – Section B. The geometric mean of the comparison values for each criterion is calculated.

$$\tilde{r}_i = \left(\prod_{j=1}^n \tilde{m}_{ij} \right)^{1/n} \quad i = 1, 2, \dots, n \quad (9)$$

5. Priority synthesis and defuzzification:

- a. The weights of the criteria are determined by normalizing the values obtained in the matrix $\tilde{M}$ as is shown in Figure 2 – Section C.

$$w_i = \tilde{r}_i \otimes (\tilde{r}_1 \oplus \tilde{r}_2 \oplus \dots \oplus \tilde{r}_n)^{-1} \quad (10)$$

- b. Each weight is a triangular fuzzy number that needs to be defuzzified.
- c. A “defuzzification” process is performed using the center of area method to convert these fuzzy numbers into scalar values b_i as is shown in Figure 2 – Section D.

$$b_i = \frac{lw_i + mw_i + uw_i}{3} \quad (11)$$

6. Normalization and selection of the best alternative:

- a. The “defuzzified” weights are normalized to ensure that the sum of all weights equals one. The alternative with the highest value $\widehat{w}_i$ is the best of all the alternatives.

$$\widehat{w}_i = \frac{b_i}{\sum_{i=1}^n b_i} \quad (12)$$

Once the normalized relative weights are established for the criteria, the same approach is employed to determine the corresponding values for the alternatives. Nonetheless, in this phase, each alternative should be compared pairwise for each of the criteria (Lee, 2014; Fu et al., 2023; Boonmee & Thoenburin, 2024).

Figure 2 synthesizes the Fuzzy AHP methodology described and illustrates how its main components are logically connected. Section A represents the hierarchical structuring of the decision problem, where the objective of selecting suitable street market locations is decomposed into evaluation criteria and candidate locations, as introduced in Sections 3.2 and 3.3. Section B corresponds to the construction of fuzzy pairwise comparison matrices and the calculation of fuzzy weights using triangular fuzzy numbers, reflecting the relative importance of criteria under uncertainty. Section C shows the normalization process, where the aggregated fuzzy weights are combined and scaled to ensure comparability across criteria. Finally, Section D illustrates the defuzzification step, in which fuzzy weights are transformed into crisp scalar values and validated through a consistency check. The resulting normalized weights quantify the qualitative suitability of each location and constitute a direct input to the optimization model, linking expert-based urban assessments with the quantitative demand allocation framework.

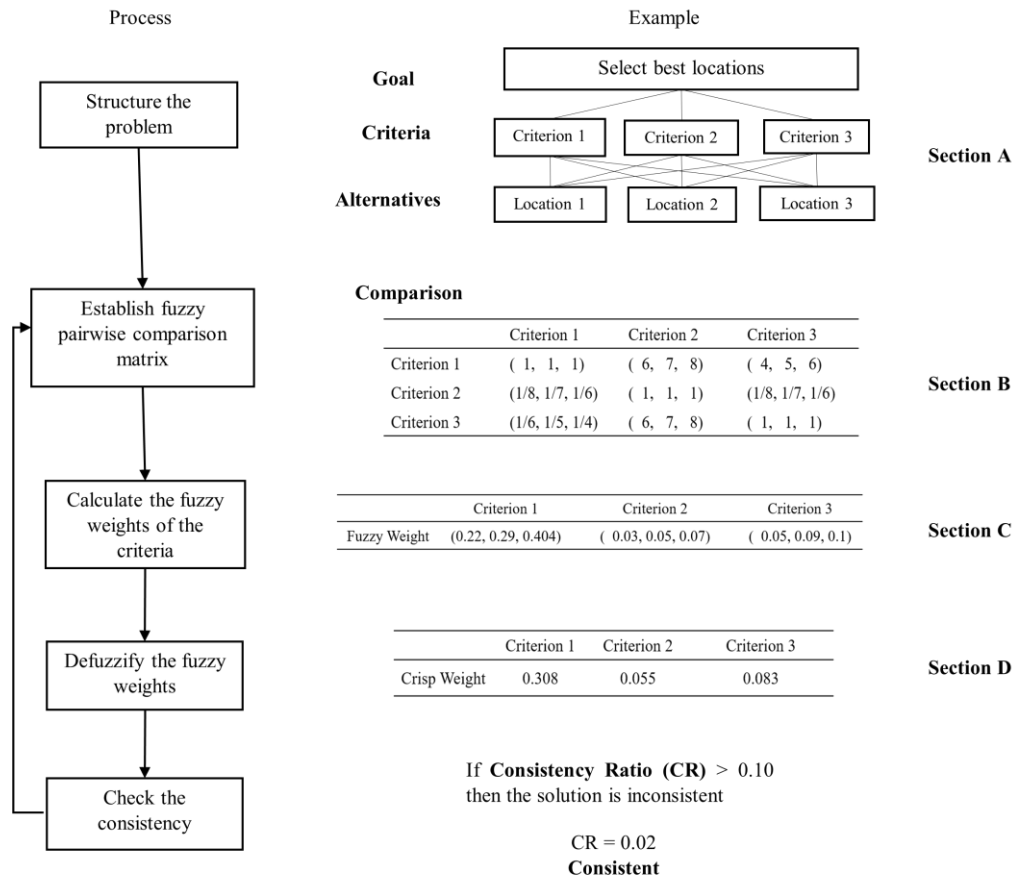


Figure 2. Process of Fuzzy AHP.

4. Methodological Framework

4.1 Data preparation

For the purpose of this research, the area of Bosa was divided into 271 300×300-meter cells. An example of this division is shown in Figure 3. This process was carried out using official geospatial data, delimiting Bosa within a Geographic Information System (GIS). We used the Python libraries GeoPandas (Jordahl, n.d.) and Folium (python-visualization, n.d.) to generate a grid of cells. Each cell was assigned a unique identifier, and its center coordinates (latitude and longitude) were stored. Each cell comprises an average of 600 residents, calculated using the population density of each zonal planning unit in Bosa as is shown in Figure 4. The average per-person demand for fruits and vegetables was set at 175.2 grams per day according to the figures of the Ministerio de Salud y Protección Social and FAO (2013). Hence, the average cell daily demand was around 840 Kg.

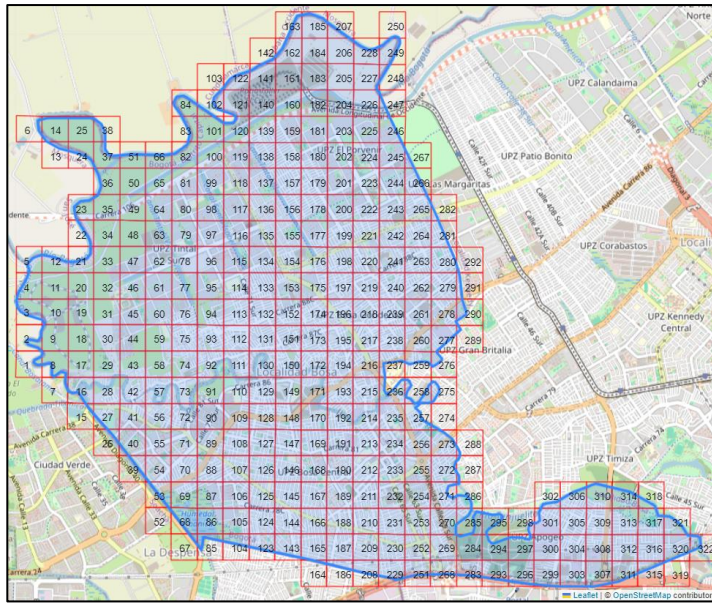


Figure 3. Division of Bosa by cells.

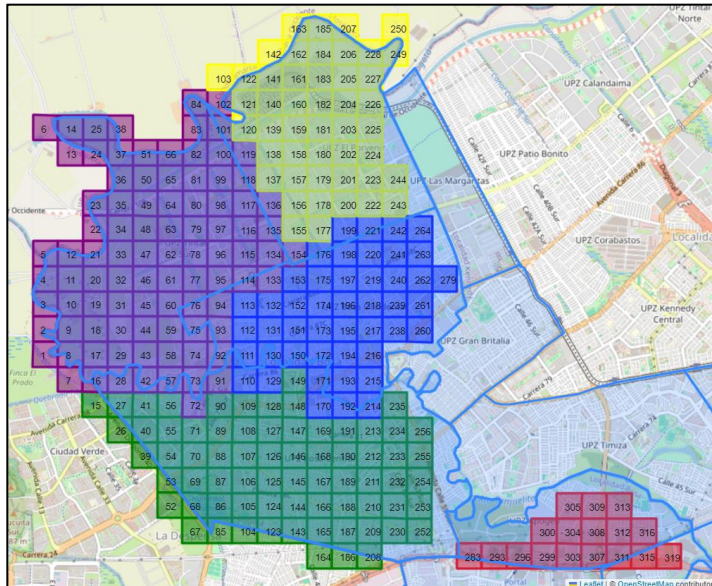


Figure 4. Division of Bosa by cells and zonal planning unit.

We identified 25 potential locations for street markets based on data retrieved from a public areas database, using available space area as criterion. Each potential location was selected by satellite imagery from Google Earth™ and cartographic data on land use extracted from Bogotá's Land Use Plan (POT in Spanish) (Alcaldía Mayor de Bogotá, 2004). These candidate locations were validated by the administration of the “Mercados Campesinos” (farmer’s markets) of the SDDE. Also, 44 large retailers were identified using Google Maps™. These retailers included “fruvers” (specialized fruit and vegetable stores), and supermarkets. Smaller stores were grouped as a single competitor category, aggregating their demand under a classification labeled as “others.”

To represent the option of shopping at other retailers, such as markets at other districts or nano stores, a 70th location was added, with at least one of these stores located within a 500-meter radius of each customer cluster and associated with higher prices to penalize the discrete choice model. To calculate distances between customers and all food seller locations, we used walking distance on actual pedestrian road networks using OpenStreetMap (OpenStreetMap contributors, n.d.). Based on Mejía et al. (2023), the weight coefficients of the logit function were established, which were then applied in a utility function. Finally, a buffer ring, a designated area surrounding

a point to measure reach or accessibility, was used to establish the maximum distance a customer would travel to a food seller, if customers only visit stores located within a radius of 750 meters or less. This is the concept of buffer ring.

4.2 Evaluation of potential locations

The qualitative evaluation of the potential locations for the street markets encompassed several factors that were analyzed to determine the appropriateness of public spaces for accommodating street markets. The set of evaluation factors was derived primarily from the specialized literature on market planning, retail location, and urban public space. Studies on retail agglomerations and marketplaces (Teller & Elms, 2010; Morales, 2011) identify accessibility, pedestrian infrastructure, and the compatibility of surrounding land uses as key determinants of market attractiveness and performance; recent assessments of urban space quality employ population density, transport accessibility, and the condition of surrounding buildings as measurable dimensions of site suitability (Li et al., 2025); and the FAO planning and design manual for markets (Tracey-White, 1991) explicitly lists site extension and shape, road access, proximity to public transport routes, and the availability of public services (water, electricity, and sanitation) as essential requirements for market sites. Analogous sets of criteria are recurrent in facility location studies that combine multi-criteria analysis with GIS or optimization approaches (Yang & Lee, 1997; Kahraman et al., 2010; Kim & Hwang, 2018). The resulting list of factors was subsequently reviewed and validated by a senior consultant in urban planning, who holds graduate studies in urban planning and has participated in relevant large-scale projects commissioned by the mayor's office of Bogotá. Her role was to confirm the relevance and applicability of the literature-based factors in the specific regulatory and urban context of Bogotá, and to verify that no locally critical factor was missing, rather than to define the evaluation criteria herself.

Based on the literature, the expert validation, and the information that could be objectively collected, the following ten factors were included in the model:

- Population density: the number of residents within a defined area.
- Stratum: social stratification based on median customer income, which affects public utility rates.
- Roads: the condition and connectivity of roads in the area.
- Sidewalks: availability and condition of pedestrian walkways.
- Morphology: the physical shape of the area of the potential location. In this context compact, square footprints work better than long, narrow ones
- Public services: The availability of essential utilities such as water, electricity, and sanitation.
- Land use: The zoning and function of surrounding areas, such as commercial, residential, or mixed-use spaces.
- Traffic: The level of vehicular congestion in the vicinity.
- Transport: Access to public transportation routes, particularly the proximity to bus stops.
- Housing: The types and quality of residential buildings in the area.

Each of these factors was analyzed within a 300-meter radius of the potential street market sites. To reduce subjectivity and make the evaluation reproducible, each factor was operationalized through an explicit indicator and measurement rule, as summarized in Table 2. Quantitative factors (e.g., population density) were measured directly on numerical scales, whereas qualitative factors (e.g., morphology, roads, sidewalks, and housing) were rated on a five-level ordinal scale (“very bad”, “bad”, “acceptable”, “good”, and “excellent”) according to predefined rules based on observable site attributes. Population estimates were derived using data from the National Administrative Department of Statistics (DANE in Spanish) (DANE, 2015; Guzman et al., 2020); road network attributes and walking distances were obtained from OpenStreetMap (OpenStreetMap contributors, n.d.); site dimensions and the condition of the physical infrastructure were assessed from Google Earth™ satellite imagery; land use classifications were taken from Bogotá's Land Use Plan (POT) (Alcaldía Mayor de Bogotá, 2004); and traffic congestion was quantified using the typical traffic layer of Google Maps™.

Table 2. Operationalization of the qualitative evaluation factors: indicator, measurement rule, and data source.

Factor	Indicator	Measurement rule and scale	Data source
Population density	Residents within the 300 m radius	Number of inhabitants estimated from official population projections at the block/UPZ level; the continuous value is used directly for ratio comparisons	DANE (2015); Guzman et al. (2020)
Stratum	Predominant socio-economic stratum	Modal stratum (1–6) of the blocks within the radius, used as an income proxy	DANE (2015); Guzman et al. (2020)
Roads	Condition and connectivity of adjacent roads	Five-level ordinal scale based on the number of paved perimeter roads and their observed surface condition (two or more paved roads in good condition = “excellent”; a single unpaved access = “very bad”)	OpenStreetMap; Google Earth™
Sidewalks	Availability and condition of pedestrian walkways	Five-level ordinal scale based on the share of the site perimeter with continuous sidewalks and their observed condition	Google Earth™ imagery
Morphology	Compactness of the usable area	Elongation ratio (shortest to longest side of the minimum bounding rectangle of the usable area): ≥ 0.75 = “excellent”; $0.50\text{--}0.75$ = “good”; $0.25\text{--}0.50$ = “acceptable”; < 0.25 = “bad”; fragmented or irregular shapes = “very bad”	Google Earth™ measurements
Public services	Availability of water, electricity, and sanitation	Ordinal scale based on the number of services available at, or immediately adjacent to, the site (3 = “excellent”; 2 = “good”; 1 = “acceptable”; 0 = “bad”/“very bad”)	Parks inventory (Alcaldía Mayor de Bogotá, 2016); POT (Alcaldía Mayor de Bogotá, 2004)
Land use	Compatibility of the surrounding zoning	Ordinal scale based on the POT classification of the surrounding areas (commercial or mixed-use = “excellent”/“good”; predominantly residential = “acceptable”; industrial or restricted uses = “bad”/“very bad”)	POT (Alcaldía Mayor de Bogotá, 2004)
Traffic	Vehicular congestion on adjacent roads	Five-level ordinal scale from the typical traffic layer during market operating hours (free flow = “excellent”; heavily congested = “very bad”)	Google Maps™
Transport	Access to public transportation	Number of bus stops within 300 m (three or more = “excellent”; two = “good”; one = “acceptable”; none = “bad”/“very bad”)	Google Maps™; OpenStreetMap
Housing	Type and quality of the surrounding residential buildings	Five-level ordinal scale based on the predominant building typology and its state of consolidation, cross-checked with the socioeconomic stratum	Google Earth™ imagery; DANE (2015)

Then the “crisp” AHP method was applied to the locations and factors described above in a typical fashion: if a criterion (say criterion i) is about as important as another (say criterion j), the cell i, j in the criteria matrix will contain the value of 1 (and the cell j, i will contain a value of 1); if criterion i is moderately more important than criterion j , the cell i, j will contain a 3 (and the cell j, i will contain a value of $1/3$) and so on. See Table 1 in Şekerci and Aydin (2022).

The relative importance of the criteria was elicited from the urban planning consultant through a structured pairwise comparison protocol: she was presented with every pair of criteria and asked to provide a comparative score on the 1–9 scale, as explained above. The role of the research team was limited to administering the

elicitation and verifying the consistency of the resulting judgments; the team did not alter the expert's preferences. AHP crisp weights and the consistency ratio (CR) were then computed. In a first pass, the CR value exceeded the acceptability threshold of 0.1; the specific inconsistent comparisons were therefore identified and returned to the consultant, who carefully re-evaluated them. The final criteria comparison matrix is consistent, with $\lambda_{\max} = 10.370$, $CI = 0.041$, and $CR = 0.028 < 0.1$ (random index $RI = 1.49$ for $n = 10$ criteria). The resulting priority ordering, which assigns the greatest importance to population density, land use, and access to public transport, is consistent with the site requirements emphasized in the market planning literature (Teller & Elms, 2010; Morales, 2011; Tracey-White, 1991), in which demand potential, compatible land use, and transport accessibility are consistently reported as first-order location factors, thereby providing external support for the elicited weights.

This evaluation process implied a three-dimensional matrix with dimensions $50 \times 50 \times 10$ (50 Locations $\times$ 50 Locations $\times$ 10 Criteria). Since direct pairwise comparisons were impractical at this scale, absolute scores were first assigned to each location on each criterion following the indicators and measurement rules defined in Table 2: quantitative criteria (e.g., population density) were scored numerically, while qualitative criteria (e.g., morphology) were rated on the five-level ordinal scale. These absolute scores were then converted into pairwise comparative scores through a fixed correspondence rule: two locations with the same rating received a value of 1, and each additional level of difference increased the intensity of preference, up to a value of 7 (and its reciprocal $1/7$) for the largest possible difference between the extreme ratings ("excellent" vs. "very bad"). For numerically scored criteria, the comparative scores were obtained by discretizing the ratio between the two locations' values onto the same scale. In this way, the location-level comparisons follow mechanically from the measured indicators rather than from case-by-case judgment.

Even with explicit measurement rules, expert-based ratings retain a degree of imprecision. To explicitly model this uncertainty and to mitigate the potential ambiguity and bias of individual judgments, we opted for the fuzzy approach: the crisp numbers were transformed into triangular fuzzy numbers, in which each single value is replaced by a range of plausible values. This methodology was also used in Ozgen and Gulsun (2014). The fuzzy scale employed is shown in Table 3.

Table 3. Fuzzy scale.

Scale	Definition	Numeric fuzzy value
1	Equal importance	(1, 1, 1)
3	Moderate importance	(2, 3, 4)
5	Strong importance	(4, 5, 6)
7	Very strong importance	(6, 7, 8)
9	Extreme importance	(9, 9, 9)

From this process, a fuzzy vector of size 50 was obtained. To defuzzify it into crisp values, which is the result of all the fuzzy AHP, this vector is represented as w_i (The weight of the location i according to qualitative criteria). This output was validated with their consistency ratio (CR); if $CR < 0.10$ the solution is consistent, else the pairwise comparisons must be reviewed. These calculations were run in python using the pyDecision library (Pereira et al., 2025).

In summary, three methodological safeguards support the validity and reproducibility of this evaluation procedure. First, the evaluation factors are grounded in the specialized literature on market planning and facility location (Yang & Lee, 1997; Kahraman et al., 2010; Kim & Hwang, 2018; Teller & Elms, 2010; Morales, 2011; Li et al., 2025; Tracey-White, 1991), and are operationalized through the explicit indicators and measurement rules of Table 2; the role of expert judgment was restricted to validating the local applicability of the factors and providing the pairwise comparison scores. Second, the consistency of the elicited judgments was formally verified through the consistency ratio, and inconsistent comparisons were re-evaluated until $CR < 0.1$. Third, the fuzzy formulation propagates the imprecision of individual judgments through the weighting process rather than treating single point estimates as exact values (Ozgen & Gulsun, 2014; Buckley, 1985). In addition, the scenario analysis presented in Section 6.1, which compares the model solutions obtained with and without the Fuzzy AHP weights,

quantifies the extent to which the weighting scheme influences the final location decisions. We acknowledge that the reliance on a single expert remains a limitation of this study; this point is further discussed in Section 7.

5. Optimization model

The optimization model formalizes the interaction between customer choice behavior, urban suitability, and operational constraints governing the deployment of street markets. Its objective is to determine which candidate locations should be activated on specific days of the week to maximize the demand captured by street markets, while respecting urban planning considerations and institutional operating rules. Customer demand is endogenously allocated through a discrete choice formulation, whereas the qualitative suitability of each location is incorporated through the weights obtained from the Fuzzy AHP analysis. In this case, we modify the traditional objective function, which is the market share maximization, with the weight obtained from the fuzzy AHP analysis. By doing so, we include the urban factors into our decision model.

Sets:

$I = \{1 \dots n\}$ Market candidates, indexed in i .

$J = \{1 \dots m\}$ Customer nodes, indexed in j .

$T = \{Monday \dots Sunday\}$ Set of days (Monday to Sunday), indexed in t .

Parameters:

w_i : Fuzzy AHP Weight for market candidate i .

d_{jt} : Demand from customer j on day t .

M_d : Maximum number of streets markets to open during the week.

vV_{ij} : Observed part of the utility perceived by customer j of alternative i .

a_{ij} : Binary parameter that takes the value of 1 if the distance from i to j is less or equal than a maximum distance (buffer ring).

Variables:

y_{ijt} : The probability of customer j of selecting street market i at period t .

x_{it} : Binary variable; 1 if the location of street market i is selected in day t ; 0 otherwise.

$$\text{Maximize } Z = \sum_{i \in I} \sum_{j \in J} \sum_{t \in T} w_i d_{jt} y_{ijt} \quad (13)$$

$$y_{ijt} = \frac{e^{v_{ij} a_{ij} x_{it}}}{\sum_{s \in I} e^{v_{sj} a_{ij} x_{st}} + \sum_{s \in S \setminus I} e^{v_{sj} a_{ij}}} \quad \forall i \in I; \forall j \in J; \forall t \in T \quad (14)$$

$$x_{it} + x_{it+1} + x_{it+2} \leq 1 \quad \forall i \in I; \forall t \in T < |T| - 2 \quad (15)$$

$$x_{i1} + x_{i2} + x_{i7} = 0 \quad \forall i \in I; \forall t \in T \quad (16)$$

$$\sum_{i \in I} \sum_{t \in T} x_{it} = M_d \quad (17)$$

$$\sum_{i \in I} x_{it} \leq 0.3 M_d \quad \forall t \in \{6, 7\} \quad (18)$$

$$\sum_{i \in I} x_{it} \leq 0.2 M_d \quad \forall t \in \{1, 2, 3, 4, 5\} \quad (19)$$

$$\sum_{i \in I} \sum_{t \in Weekdays} x_{it} = 0.4M_d \quad (20)$$

$$x_{it} \in \{0,1\} \quad \forall i \in I; \forall t \in T \quad (21)$$

$$y_{ijt} \in \{0,1\} \quad \forall i \in I; \forall j \in J; \forall t \in T \quad (22)$$

The objective function (13) maximizes the total weighted demand captured by street markets. Demand from each customer node and day is weighted by the qualitative suitability of the selected market location, represented by the Fuzzy AHP weight. This formulation ensures that locations with favorable urban conditions contribute more strongly to the objective, linking qualitative urban assessments with quantitative demand allocation. Constraint (14) models customer choice using a multinomial logit formulation, where the probability of selecting a street market depends on its perceived utility, its availability on a given day, and spatial accessibility captured through the buffer ring parameter. The inclusion of competing sellers accounts for alternative purchasing options available to customers, reflecting realistic competitive market conditions. The buffer ring assumption limits customer choice to sellers within a reasonable walking distance, consistent with empirical findings on urban food access behavior. Constraints (15) and (16) regulate the temporal continuity of market operations. They prevent the same street market from operating on three consecutive days and from opening on both the last and first days of consecutive weeks. These constraints reflect operational practices observed in Bogotá's street market program, where markets are designed as periodic events rather than permanent installations, ensuring logistical feasibility and equitable spatial rotation (Metropoli Online, n.d.; Alcaldía Mayor de Bogotá, n.d.-a). Constraint (17) fixes the total number of street markets operating during the week, representing budgetary and organizational capacity limitations faced by public authorities. Constraints (18) and (19) restrict the maximum proportion of markets that can operate on any single day, differentiating between weekdays and weekends. This structure avoids excessive concentration of markets on specific days and promotes a balanced temporal distribution. Constraint (20) enforces a minimum level of weekday operation, guaranteeing daily access to fresh food beyond weekend events and aligning with food security objectives. The numerical thresholds adopted in these constraints are based on current operational guidelines and planning practices of Bogotá's Secretariat of Economic Development.

Constraints (21) and (22) define the binary nature of the decision variables, indicating whether a market is opened and whether a customer selects a given market on a specific day.

The second constraint set involves non-linear aspects, we have adapted the linearization method proposed by Haase and Müller (2014). Consequently, this constraint set is reformulated into constraints (23) – (27), introducing the variable $\bar{y}_{jt}$. This variable accumulates the probability of customers selecting sellers during period t and employs the auxiliary parameter φ_{jt} .

$$\varphi_{ij} = \frac{e^{v_{ij}a_{ij}}}{\sum_{s \in J \setminus I} e^{v_{sj}}} \quad \forall i \in I; \forall j \in J \quad (23)$$

$$\bar{y}_{jt} + \sum_{i \in I} y_{ijt} \leq 1 \quad \forall j \in J; \forall t \in T \quad (24)$$

$$y_{ijt} - \frac{\varphi_{ij}}{1 + \varphi_{ij}} x_{it} \leq 0 \quad \forall i \in I; \forall j \in J; \forall t \in T \quad (25)$$

$$y_{ijt} - \varphi_{ij} \bar{y}_{jt} \leq 0 \quad \forall i \in I; \forall j \in J; \forall t \in T \quad (26)$$

$$\bar{y}_{jt} \in \{0,1\} \quad \forall j \in J; \forall t \in T \quad (27)$$

Parameter (23) calculates the marginal probability that customers will buy fruits and vegetables from competitor sellers, which are essentially the alternatives where they have no other choice. Constraint (24) computes the accumulated probability ($\bar{y}_{jt}$) of customers opting for sellers other than the street markets. Constraints (25) and (26) establish connections between the decision variables and an auxiliary parameter known as ϕ_{ij} . Lastly, constraint (27) sets the range for the decision variable $\bar{y}_{jt}$.

6. Computational results and analysis

The experiments were run on a computer with an AMD Ryzen 5 – 3450U microprocessor, with 8 GB RAM, and 2.1 GHz. The Fuzzy AHP analysis was performed in Python, followed by the implementation and solution of the mathematical model, also in Python, using Gurobi™. We proposed different scenarios where the parameters were varied. Our baseline scenario considered a set of ten markets throughout the week, with a demand distribution of 50% on weekdays and 50% on weekends, a buffer ring radius of 750 meters minimum. The result for the base scenario is shown in Figure 5, Figure 6 and Table 4.

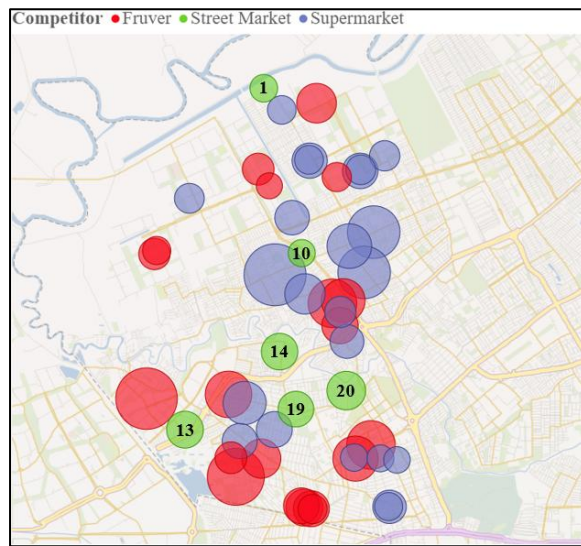


Figure 5. Results for base scenario.

Table 4. Street markets to open throughout the week.

Street Market (ST)	Mon	Tue	Wed	Thu	Fri	Sat	Sun
ST 01						X	
ST 10						X	
ST 13			X				X
ST 14				X			X
ST 19		X				X	
ST 20			X				X

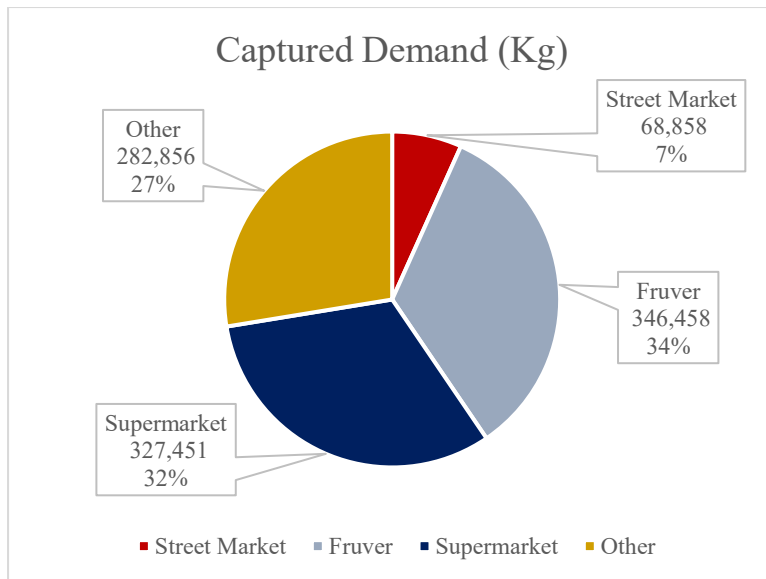


Figure 6. Capture demand per competitor type.

Figure 5 illustrates the street markets to be opened throughout the week, where the diameter of each circle represents the demand captured by the retailer. Larger circles indicate higher demand in that specific location. The color coding shows the type of competitor present. Table 4 details the street markets scheduled to open on each day of the week under the baseline scenario. The "X" marks indicate the days on which a specific market is open. For example, ST 01 is open on Saturdays, while ST 20 operates on Wednesdays and Sundays. It's clear that in the base scenario, the market distribution takes place on the weekend. This aligns with the current strategy of Bogotá's Secretariat of Economic Development, which opens the markets on Saturdays and Sundays; however, it envisions greater flexibility throughout the week (Alcaldía Mayor de Bogotá, n.d.-b; Alcaldía Mayor de Bogotá, 2025). Figure 6 presents the distribution of captured demand, measured in kilograms, among different types of retailers. Customer demand for fruits and vegetables in Colombia is primarily distributed among neighborhood stores, supermarkets, and specialized produce shops (fruvers). According to a 2022 report by the Food Industry Chamber of ANDI (National Business Association of Colombia), neighborhood stores accounted for 35% of food and beverage purchases, while supermarkets captured 28.4%. These data indicate that the proposed solution is similar to the observed market distribution trends (Encolombia, n.d.).

Beyond the numerical results, the baseline scenario provides insights relevant for urban food policy. The selected locations tend to concentrate in areas that balance demand intensity with favorable urban conditions, as reflected by the Fuzzy AHP weights. This indicates that the model does not simply prioritize densely populated areas, but rather locations where accessibility, infrastructure quality, and surrounding land use jointly support market operation. From a policy perspective, this suggests that street markets can be strategically deployed to complement existing retail structures rather than directly competing with them, reinforcing access to fresh food in areas that are both socially and logistically suitable. The observed concentration of market activity on weekends reflects current institutional practice, yet the model also identifies feasible weekday locations, supporting the possibility of extending market operations beyond traditional schedules without compromising efficiency.

6.1 Impact of fuzzy AHP and buffer ring

We modified the radius of the buffer ring to a maximum distance of 1,000, 1,500 and 2,000 meters to explore its influence in the market distribution. We also compared the models using the weights derived from Fuzzy AHP (consideration of urban factors) vs No Fuzzy AHP weights (without considering urban factors). This is conducted to assess how the inclusion of urban factors influences distribution of demand. This analysis aims to determine whether considering these factors leads to better-informed decisions regarding market distribution, beyond the

absolute amount of captured demand. Table 5 and Figure 7 illustrate how the captured demand varies with those considerations.

Table 5. Scenarios with Fuzzy AHP.

Scenarios	Buffer Ring	Captured demand with fuzzy AHP weights (Kg)	Captured demand without fuzzy AHP weights (Kg)
Base	750 m	68,858	69,046
Scenario 1	1,000 m	72,595	75,450
Scenario 2	1,500 m	78,159	84,342
Scenario 3	2,000 m	85,025	93,085

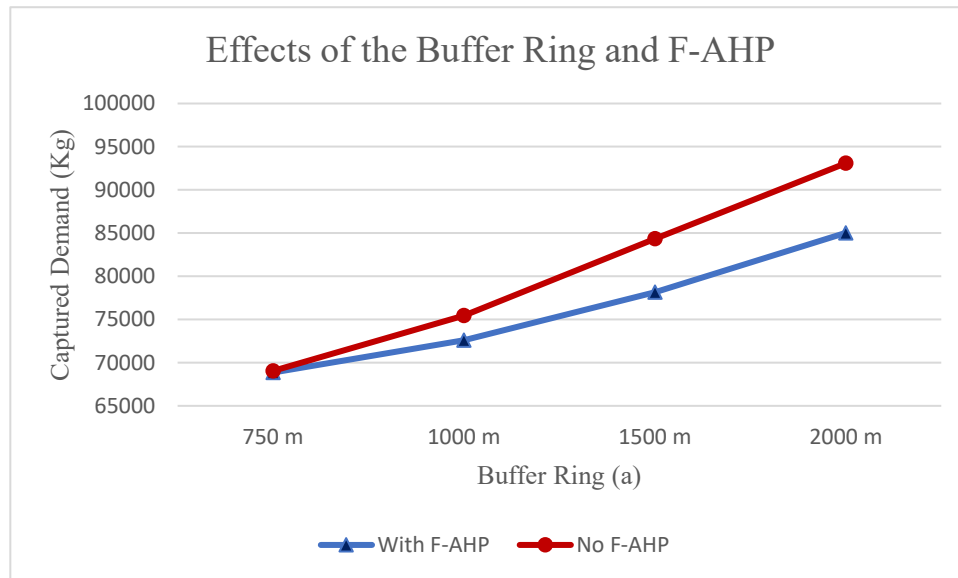


Figure 7. Effects of the buffer ring and fuzzy AHP.

The exclusion of Fuzzy AHP shows an increase in captured demand, as illustrated in Figure 7. This result indicates that without Fuzzy AHP, the model estimates higher demand capture due to the lack of qualitative urban factors. Without Fuzzy AHP, the model tends to maximize captured demand without considering whether markets are in the most suitable areas from an urban and social perspective. Fuzzy AHP weights integrate essential urban factors such as accessibility, infrastructure, and mobility patterns. By doing so, it provides a more realistic perspective of market behavior, which helps in identifying locations that are not only profitable in the short term but also sustainable over time. Without these considerations, the model might prioritize areas with high immediate demand but overlook potential long-term inefficiencies. Ultimately, the comparison demonstrates the importance of incorporating urban factors into the decision-making process. The model with Fuzzy AHP weights may show slightly lower demand, yet it ensures a more balanced and accessible distribution, which is essential for sustainable growth and market efficiency. Finally, increasing the buffer ring size increases the demand of capture by enabling markets to reach more distant customers, possibly tapping into areas with lower market saturation.

The comparison between scenarios with and without Fuzzy AHP highlights why certain locations are selected over others. When qualitative urban factors are excluded, the model favors locations with high immediate demand capture, even if they exhibit limited accessibility or weaker urban infrastructure. In contrast, the inclusion of Fuzzy AHP weights shifts the solution toward locations with greater long-term suitability, such as improved pedestrian access, proximity to public transport, and compatible land use. As a result, some high-demand locations are intentionally deprioritized in favor of sites that better support sustained market operation and equitable access. This trade-off explains the slightly lower captured demand observed in the Fuzzy AHP scenarios and reinforces the role of urban criteria as a corrective mechanism rather than a purely restrictive one.

6.2 Number of markets

The number of markets also varied between the scenarios with 5, 10 and 15 markets to assess scale effects as is shown in Table 6.

Table 6. Variation for numbers of street markets.

Scenarios	Number of street markets	Captures demand (Kg per week)
Base	5	34,569
Scenario 1	10	68,858
Scenario 2	15	89,942

The data shows an approximate linear relationship between the number of markets and the captured demand. Using the points from the table, this relationship can be expressed by the equation $y = 7,343.1 * x - 8,974.7$, where y represents the captured demand in kilograms, and x is the number of markets. With only 5 markets, the demand captured is 34,569 kg, whereas with 15 markets, it rises to 89,942 kg, indicating that a larger number of markets substantially increases the ability to serve the customer base. Running fifteen markets per week is beyond the current SDDE capability but an increase in the budget for this initiative is projected.

6.3 Demand distribution

In addition, we adjusted demand distribution patterns, analyzing scenarios with 60% demand concentrated on weekends and 40% on weekdays, and an equal distribution throughout the whole week. Table 7 summarizes the results, showing a significant increase in captured demand when more demand is concentrated on weekends. Specifically, Scenario 1 (40% weekdays – 60% weekends) resulted in 76,059 kg of captured demand. Scenario 2 (Constant demand) achieved 54,651 kg.

Table 7. Variation for demand.

Scenarios	Demand Distribution	Captured Demand (Kg)
Base	50% Weekdays - 50% Weekends	68,858
Scenario 1	40% Weekdays - 60% Weekends	76,059
Scenario 2	Equal demand each day	54,651

These results highlight the critical importance of aligning operations with peak demand periods, particularly at weekends. This finding underscores the need for strategic planning in the scheduling and allocation of resources, focusing on periods of high customer activity.

6.4 Sensitivity and robustness discussion

The scenario analysis conducted in Sections 6.1–6.3 also serves as a sensitivity assessment of the proposed model. Variations in the buffer ring radius, number of markets, and temporal demand distribution consistently produce intuitive and stable trends. Increasing the buffer ring expands customer reach and systematically increases captured demand, while larger numbers of markets yield an approximately linear growth in demand served. Similarly, reallocating demand toward weekends amplifies captured demand, reflecting empirically observed shopping behavior. Importantly, across all scenarios, the relative influence of Fuzzy AHP remains consistent: solutions that incorporate urban factors prioritize spatial balance and accessibility over purely demand-driven outcomes. These stable patterns indicate that the model is robust to reasonable parameter variations and that its qualitative conclusions are not driven by a single calibrated scenario.

Regarding the buffer radius, the results show that increasing the maximum walking distance systematically expands the potential customer base and increases captured demand across all scenarios. Importantly, the relative ranking of selected locations remains stable, indicating that the model is not overly sensitive to moderate changes

in this parameter. This suggests that the buffer radius primarily affects the scale of demand capture rather than the structural configuration of market locations.

The comparison between scenarios with and without Fuzzy AHP weights highlights a different type of sensitivity. Excluding qualitative urban criteria leads to higher short-term demand capture but shifts the solution toward locations with weaker accessibility or infrastructure conditions. When Fuzzy AHP weights are incorporated, the model consistently favors more balanced and accessible locations, even at the expense of a marginal loss in demand. This behavior demonstrates that Fuzzy AHP functions as a corrective mechanism rather than a restrictive one, thereby enhancing the robustness of location decisions from managerial and policy perspectives.

6.5 Managerial insights

Several managerial and operational implications can be derived from this research. Currently, operating these markets in Bogotá entails an average cost of COP 4 million (approx. USD 1,000) per event, covering logistics, transportation, security, and setup, according to the Secretaría de Desarrollo Económico. The current strategy focuses on periodic events organized in under-supplied areas with high potential customer flow, using accessibility criteria and urban planning regulations. Given a weekly schedule with 10 markets operating, the total cost would amount to COP 40 million per week, considering the average cost per event for one of 20 districts. This represents a significant expense, which becomes unsustainable when weighed against the limited demand these markets can capture. From an economic perspective, this raises concerns about the long-term viability of the current strategy.

In addition to cost and demand considerations, the integration of cultural and creative dimensions into street markets could enhance their long-term sustainability. International frameworks emphasize that cultural and creative industries act as drivers of social cohesion, urban regeneration, and local competitiveness, especially when aligned with community-based events and creative districts (UNESCO & The World Bank, 2021; OECD, 2022). In Colombia, the so-called Orange Economy policy has shown how creative clusters and cultural programming generate spillovers in employment, social inclusion, and place-making, turning markets into engines of local development (Encolombia, n.d.). Similarly, research on urban culture and street vending highlights that public trust, low-income consumption patterns, and cultural identity often sustain these markets beyond purely economic rationales (Al-Jundi et al., 2022). For Bogotá, it may be valuable to explore ways to link street markets with district-led cultural agendas such as festivals (Secretaría de Cultura, Recreación y Deporte, n.d.) or heritage promotion (Alcaldía Mayor de Bogotá, n.d.-c), and possibly even to integrate them with other types of markets like those dedicated to crafts. Such strategies could help reduce their vulnerability to low demand, increase citizen participation, and reposition them as hybrid spaces of commerce and culture.

A final thought on technology: Digital transformation can significantly enhance the planning and management of street markets in emerging countries by improving data availability, coordination, and decision-making across the food system. The use of digital tools such as mobile applications, geospatial platforms, and data analytics can potentially help authorities and market managers to better understand demand patterns, vendor locations, product flows, and consumer accessibility. These technologies may support location planning, scheduling, and allocation of space, while improving transparency, traceability, and compliance with health and safety standards. In addition, digital payment systems and information-sharing platforms can formalize the integration of street markets with formal supply chains, reduce transaction costs, increase resilience, ultimately contributing to more inclusive, efficient, and sustainable urban food systems.

7. Conclusions and future work

This work contributes to the initiatives of street markets by offering an integrated approach to the location of street markets, utilizing Mixed Integer Non-Linear Programming (MINLP) and Fuzzy AHP to optimize both qualitative and quantitative factors.

It is evident that only a small portion of the demand is captured by the street markets. This result may be influenced by how the utility perceived by each customer is estimated, specifically, how the probability is

calculated using the Logit model. With a large number of competitors, the Logit model tends to favor alternatives like fruvers and supermarkets. A more robust estimator for the probabilities could improve the calculation of perceived utility, due to the limited availability of data and time, more robust models were not implemented.

A further limitation concerns the qualitative evaluation of the candidate locations. Although the evaluation factors were drawn from the market planning and facility location literature and operationalized through explicit indicators and measurement rules, the pairwise judgments used to derive the criteria weights were elicited from a single urban planning expert. The subjectivity associated with this elicitation was mitigated through formal consistency verification ($CR = 0.028$), the use of triangular fuzzy numbers to model judgment imprecision, and the sensitivity analysis of Section 6.1, which bounds the effect of the weighting scheme on the resulting solutions. Nevertheless, the weights ultimately reflect a single professional perspective. Future research should therefore involve a panel of experts and stakeholders, including planners, vendors, and community representatives, aggregate their judgments through group AHP or Delphi-based procedures, and contrast the resulting weights with revealed-preference evidence obtained from the observed performance of operating markets.

For future work, we identify the need for a more detailed approach to street market location analysis, which could involve the development of specialized markets catering to specific nutritional requirements, such as vitamins, proteins, etc. Additionally, future research could explore operational factors such as the capacity of markets and competitors, market sustainability, or the creation of specialized markets that go beyond fruits and vegetables, including other types of food like fish, meat, and more. This approach can align market locations with customer nutritional needs, health trends, dietary restrictions, and cultural preferences, considering the evolving behavior of consumers. These aspects were not included in the current study due to its primary focus on general demand patterns and spatial distribution. The scope of this research was limited to broader market dynamics, without going into specific nutritional or operational factors.

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Abbreviations

The following abbreviations are used in this manuscript:

MINLP	Mixed Integer Non-Linear Programming
AHP	Analytic Hierarchy Process
SDDE	Secretaría Distrital de Desarrollo Económico
TFN	Triangular Fuzzy Numbers
FLP	Facility Location Problem
CFLP	Competitive Facility Location Problem
MILP	Mixed Integer Non-Linear Programming
GRASP	Greedy Randomized Adaptive Search Procedure
QFD	Quality Function Deployment
ANP	Analytic Network Process
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
MOPLP	Multi-Objective Possibilistic Linear Programming
GIS	Georeferenced Information System

MCDM	Multiple Criteria Decision-Making
DANE	Administrative Department of Statistics
CR	Consistency Ratio
ST	Street Markets
ANDI	National Business Association of Colombia

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